DATA CENTERS AND ELECTRICITY

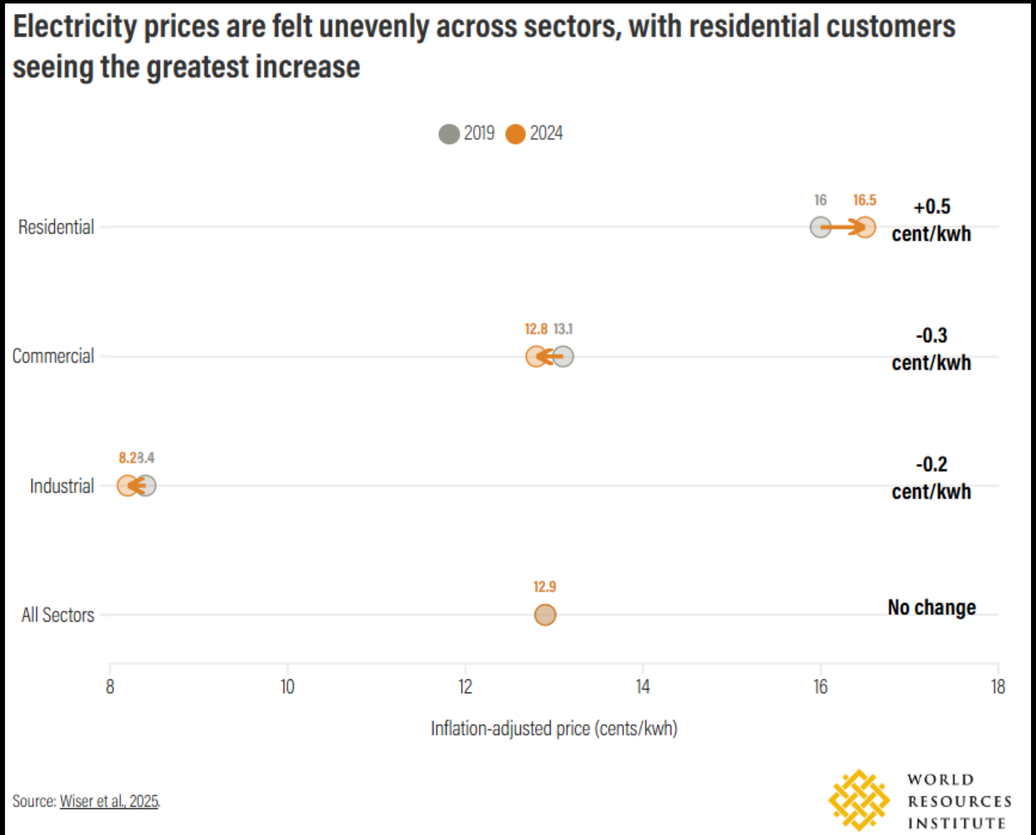
Joshua Mark and
Henry Swiber

BACKGROUND

ELECTRICITY PRICES

A COMPLICATED STORY

- When adjusted for inflation, electricity rates have actually gone down / stayed the same
- Residential rates have increased, while industry and commercial rates have fallen
- This means residential customers may be getting hit harder than commercial and industrial entities.



ELECTRICITY PRICES

DRIVERS OF COST

- Most experts do believe data centers do impact electricity cost.
- Data centers often require upgrades to local infrastructure to match level of demand.
- There are many other factors:
  - Generally aging infrastructure
  - Severe weather events
  - Shifting energy sources

DATA CENTERS

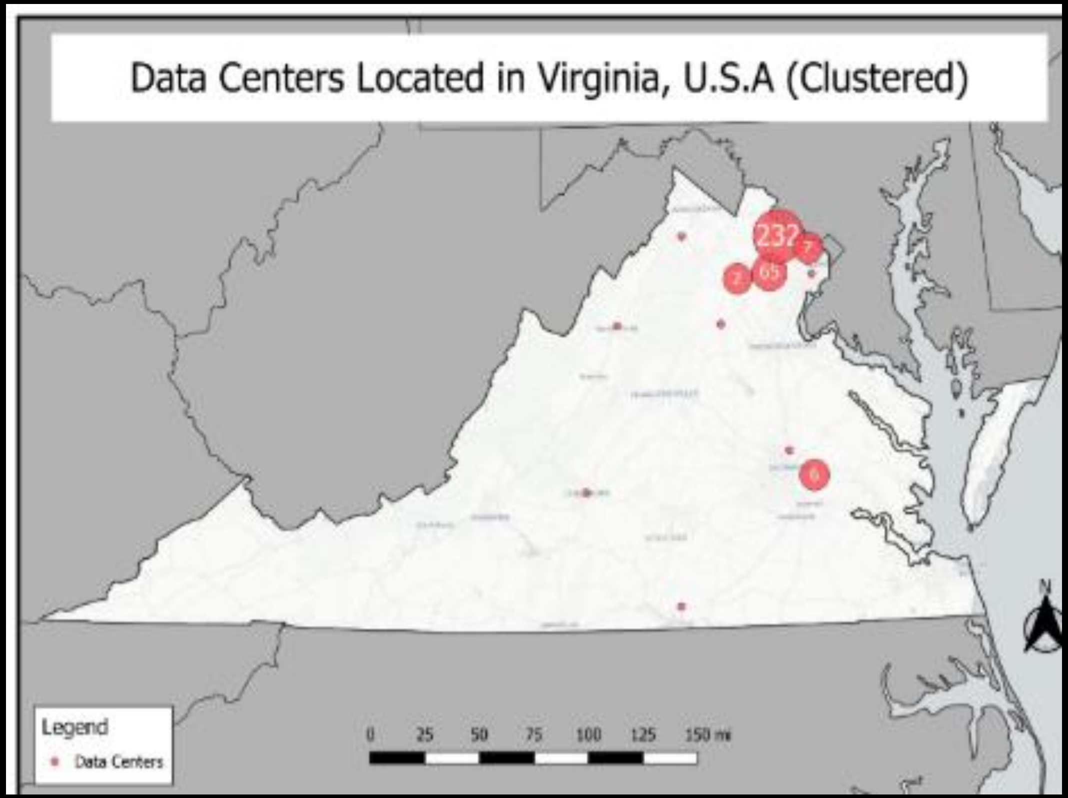
WHAT ARE THEY?

- Complexes used to house data infrastructure
  - Servers, storage, networking capabilities, etc
- Recent development: hyperscale data centers
  - Uses electricity at the scale of small towns
- Requires a large amount of electricity to fuel intense computation requirements.
- As Artificial Intelligence is growing in popularity, data centers require more processing power and thus need more electricity.

DATA CENTERS

ROLE OF VIRGINIA – A Hotspot

- Most experts do believe data centers do impact electricity cost.
- Data centers often require upgrades to local infrastructure to match level of demand.
- There are many other factors:
  - Generally aging infrastructure
  - Severe weather events
  - Shifting energy sources



THE DATA

ZIP CODES + ELECTRICITY

ELECTRICITY RATE DATA

- U.S. Electric Utility Companies and Rates with zip codes. (2014 and 2024). Provided by OpenEI and authored by Jay Huggins of NREL.
- Has electricity rate data for almost all zip codes in the United States from 2013 to 2024. For each year, there are 2 datasets: IOU and NON-IOU.
  - IOU: Investor-owned Utility
  - NON IOU: Non-Investor-owned Utility
- In each dataset, Electricity Rates are split into 3 categories:
  - comm_rate: The rate for Commercial Entities.
  - ind_rate: The rate for Industrial Entities.
  - res_rate: The rate for Residential Entities.

ZIP CODE SHAPEFILE

- 2024 Zip code boundaries provided by Esri themselves.
- Has approximately all zip codes in the United States, with each zip code being its own polygon.
- What we used:
  - Zip Code and associated Polygon

Huggins, J. (2025). U.S. Electric Utility Companies and Rates: Look-up by Zip Code (2014-2024). [Data set]. Open Energy Data Initiative (OEDI). National Renewable Energy Laboratory (NREL). https://data.openei.org/submissions/8563

esri_dm. (2026). *ArcGIS*. Arcgis.com; ESRI. https://www.arcgis.com/home/item.html?id=5f31109b46d541da86119bd4cf213848

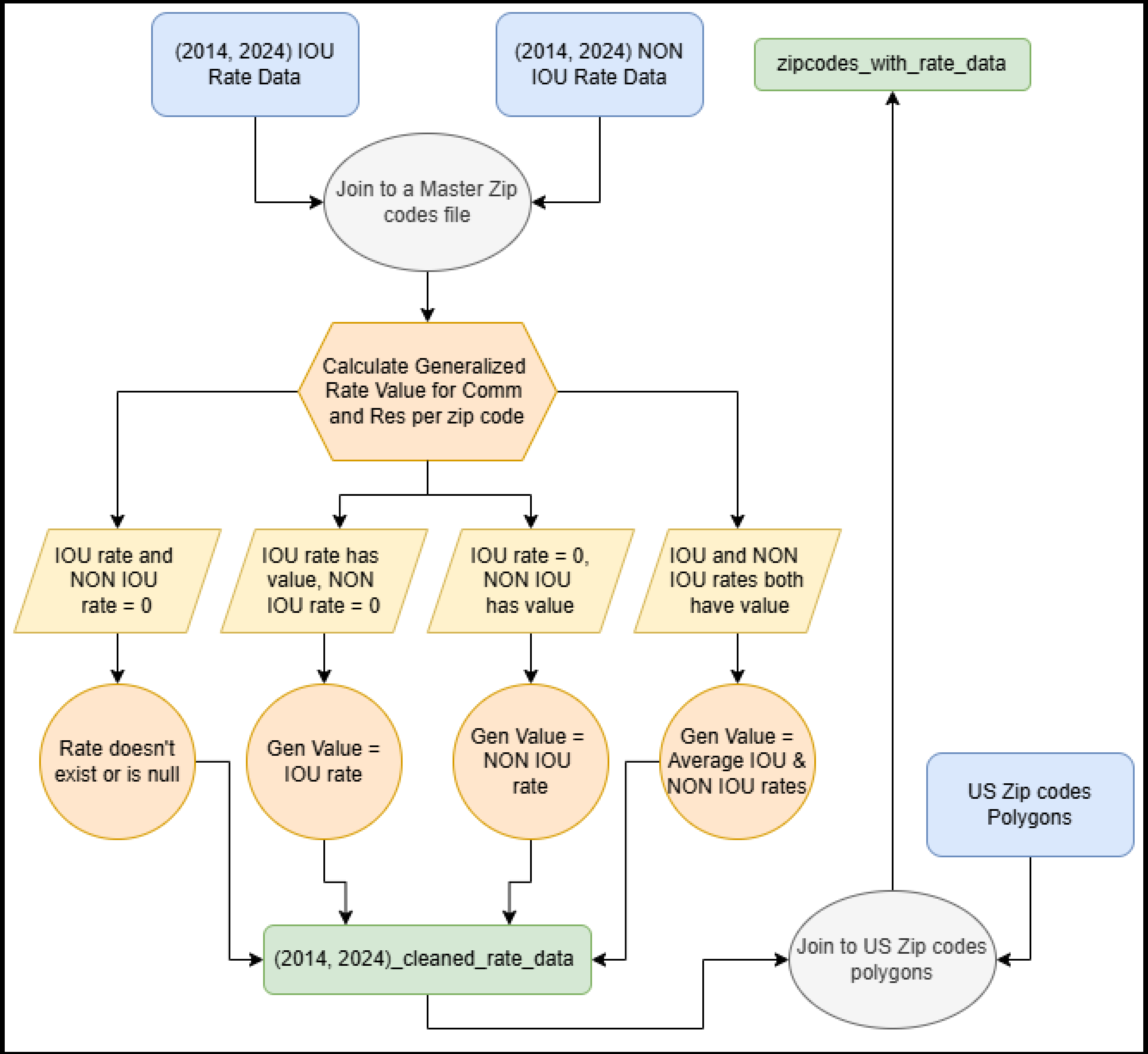
DATA CENTER LOCATIONS

- IM3 Open Source Data Center Atlas provides the locations of data centers in the United States. Locations are points with coordinates.
- Locations are sourced from OpenStreetMap and then cleaned.
- Due to the open-source nature of the data, naturally, the quantity of data centers is lower than in privately owned sources.
- Open-source data was used because private entities can charge between hundreds and tens of thousands of dollars for datasets.

Mongird, K., Vernon, C., Burleyson, C., Akdemir, K. Z., & Rice, J. (2026). IM3 Open Source Data Center Atlas (v2026.02.09) [Data set]. MSD-LIVE Data Repository. https://doi.org/10.57931/3017294

PREPARING THE DATA

- The master zip codes file contains all zip codes that are present in the IOU and NON IOU datasets.
  - Both sets cover different zip codes but have a few overlaps.
- As there is overlapping data for zip codes, logic had to be enforced to create a generalized rate value for both res and comm.
  - Both datasets were utilized to supplement gaps in the other to create a full zip code dataset.
  - When there were values from both iou and non iou, then combine and average the two values.
- This process was run through for 2014 and 2024 iou and non iou data.
  - Both 2014 and 2024 cleaned rate data were joined to the us zip codes polygons dataset

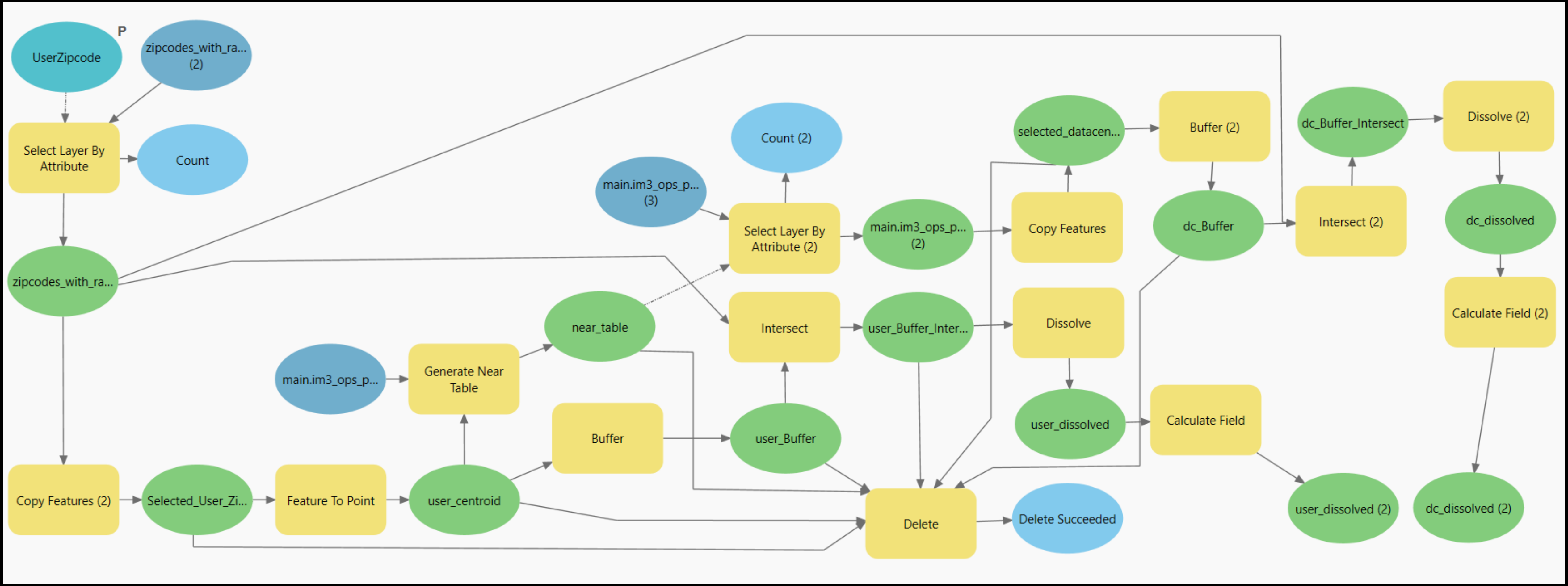


METHODOLOGY

THE METHODOLOGY

Details:

- Process starts with the user's inputted zip code
- The two files that contain data to give to the user:
  - user_dissolved and dc_dissolved
- After the process is completed, delete all unnecessary files generated.

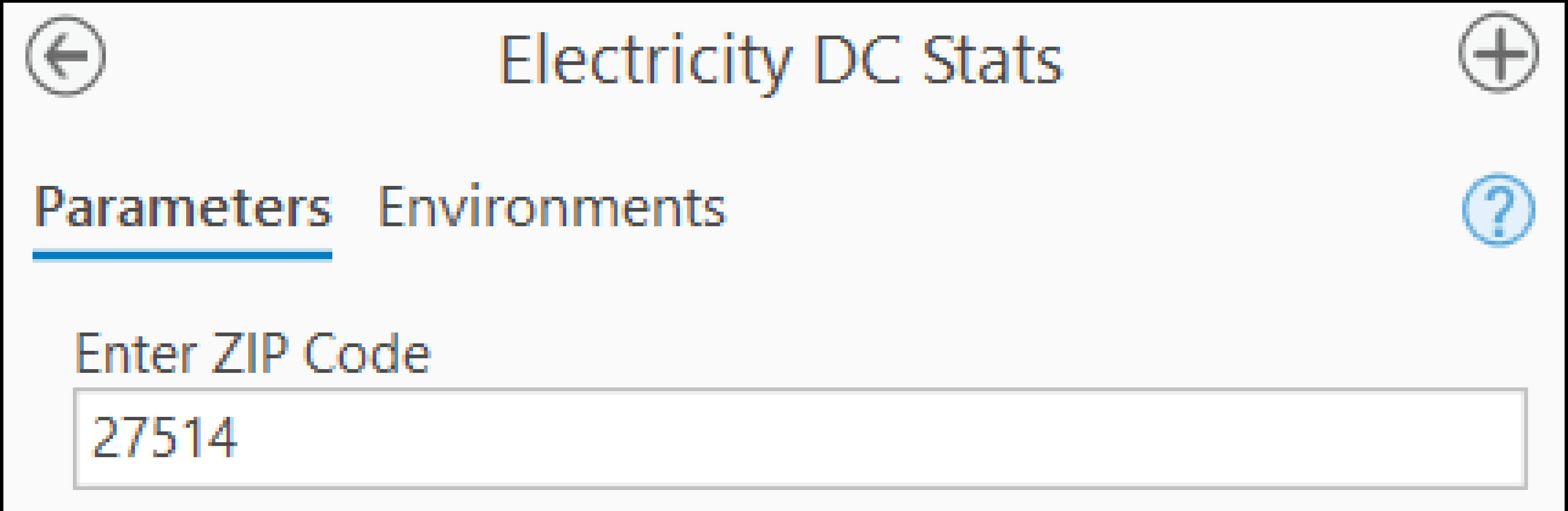


TRANSITION INTO USABLE TOOL

- Exported from ModelBuilder into a Python toolbox script
- Added a weighted by area feature when calculating average rate for the buffer areas.
    - Rates are weighted based on the area they take up within 20-mile buffer
- Add ZIP Code parameter user input
    - Error check: must be 5 digits.

```
class elecDCstats:
    def __init__(self):
        """Define the tool (tool name is the name of the class)."""
        self.label = "Electricity DC Stats"
        self.description = "This tool allows the user to input their zi

    def getParameterInfo(self):
        """Define the tool parameters."""
        param0 = arcpy.Parameter(displayName="Enter ZIP Code", name="zc

        params = [param0]
        return params

    def isLicensed(self):
        """Set whether the tool is licensed to execute."""
        return True

    def updateParameters(self, parameters):
        """Modify the values and properties of parameters before intern
        validation is performed.  This method is called whenever a para
        has been changed."""
        zip_code = parameters[0]
        # Check that input is 5 digits long
        if zip_code.value:
            import re
            if not re.match(r'^\d{5}$', zip_code.value):
                zip_code.setErrorMessage("Enter a valid ZIP code (e.g.,
        return

    def updateMessages(self, parameters):
        """Modify the messages created by internal validation for each
        parameter. This method is called after internal validation."""
        return

    def execute(self, parameters, messages):
        """The source code of the tool."""

        zipcode_input = parameters[0].valueAsText
        zipcodes = "zipcodes_with_rate_data"
        data_centers = "main.im3_ops_points"
        near_dist = None
```

ANALYSIS



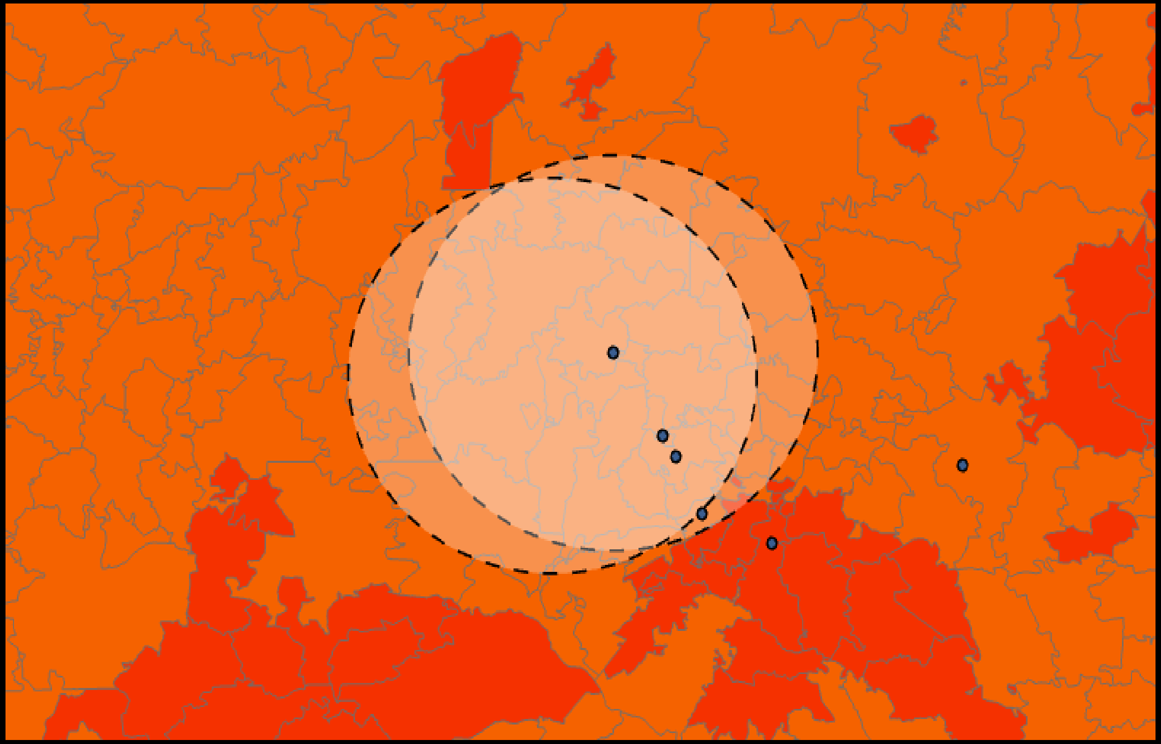
THE TOOL

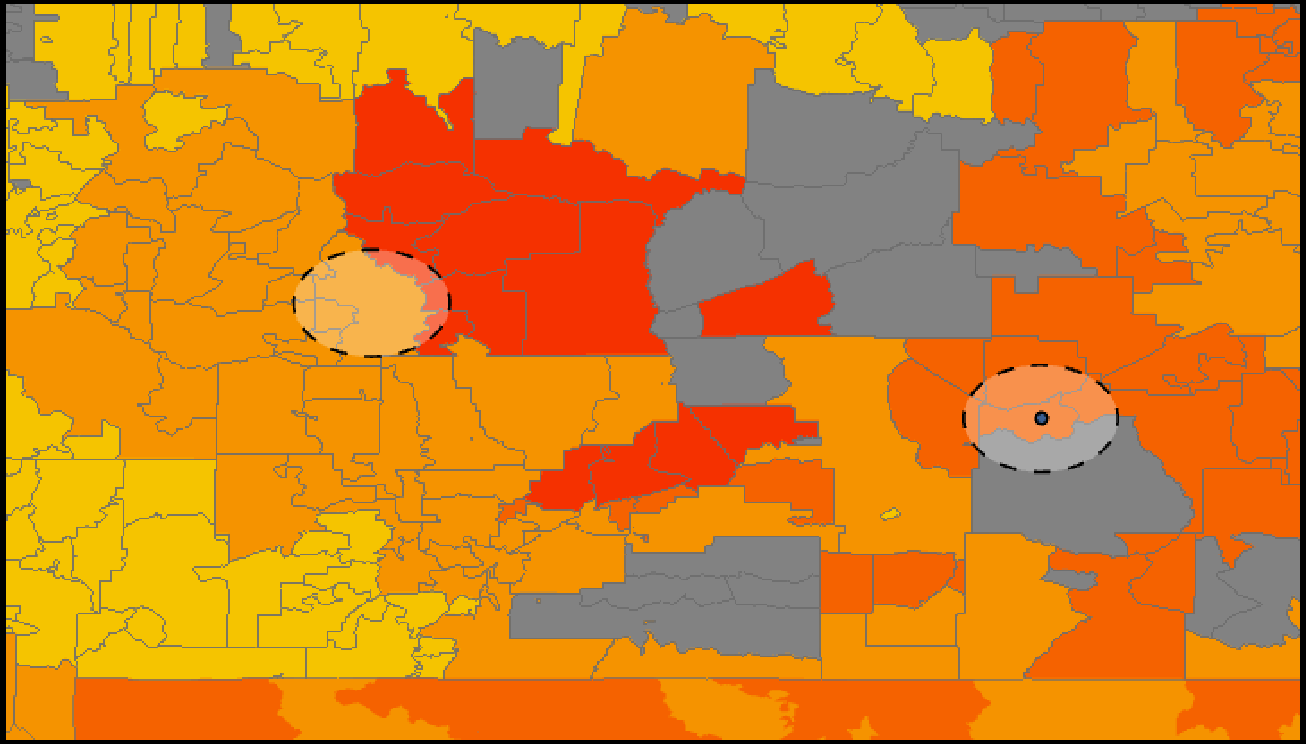
- Text input for zip code
- Will check to make sure input is 5 digits.

OUR RESULTS

```
--- User ZIP 27514 ---
Avg residential rate 2014: 0.1200 cents/kWh
Avg residential rate 2024: 0.1480 cents/kWh
Percent change: 23.33%
Distance to nearest datacenter: 0.11 miles
--- Nearest Datacenter ---
Avg commercial rate 2014: 0.0959 cents/kWh
Avg commercial rate 2024: 0.1118 cents/kWh
Percent change: 16.59%
```

- For the example:
  - The User inputted the zip code: 27514





```
--- User ZIP 27514 ---
Avg residential rate 2014: 0.1200 cents/kWh
Avg residential rate 2024: 0.1480 cents/kWh
Percent change: 23.33%
Distance to nearest datacenter: 6.36 miles
--- Nearest Datacenter ---
Avg commercial rate 2014: 0.0959 cents/kWh
Avg commercial rate 2024: 0.1118 cents/kWh
Percent change: 16.59%
```

```
--- User ZIP 59457 ---
Avg residential rate 2014: 0.1060 cents/kWh
Avg residential rate 2024: 0.1080 cents/kWh
Percent change: 1.89%
Distance to nearest datacenter: 177.50 miles
--- Nearest Datacenter ---
Avg commercial rate 2014: 0.1073 cents/kWh
Avg commercial rate 2024: 0.0754 cents/kWh
Percent change: -29.71%
```

CONCLUSION

OUR CONCLUSION

In our findings, throughout the creation of this tool, residential areas have experienced a higher rate increase than commercial areas, which have increased by around half of the residential increase. Some areas, due to a lack of data centers, proximity, or another unknown factor, have experienced very little change in electricity rate. However, there are many factors that affect electricity rates, and proximity to a datacenter is but a small spark in the sea of flames that is how electricity rates are priced.

Datacenters have many factors that can affect surrounding areas, such as aging infrastructure, companies restricting power, and insufficient power generation to support computation. Our tool allows analysis of the surrounding area of a user’s given zip code and the nearest data center. This allows the user to reach a conclusion about how at risk their area's electricity rate is from being affected by data centers.

REFERENCES

- esri_dm. (2026). *ArcGIS*. Arcgis.com; ESRI. https://www.arcgis.com/home/item.html?id=5f31109b46d541da86119bd4cf213848
- Greenfield, N. (2026, March 20). AI Data Centers: Big Tech's Impact on Electric Bills, Water, and More. Consumer Reports. https://www.consumerreports.org/data-centers/ai-data-centers-impact-on-electric-bills-water-and-more-a1040338678/
- Huggins, J., & Laboratory, N. (2015). *U.S. Electric Utility Companies and Rates: Look-up by Zipcode (2014)*. [Data set]. Open Energy Data Initiative (OEDI). National Renewable Energy Laboratory. https://data.openei.org/submissions/445
- Huggins, J. (2025). U.S. Electric Utility Companies and Rates: Look-up by Zip Code (2024). [Data set]. Open Energy Data Initiative (OEDI). National Renewable Energy Laboratory (NREL). https://data.openei.org/submissions/8563
- Mongird, K., Vernon, C., Burleyson, C., Akdemir, K. Z., & Rice, J. (2026). IM3 Open Source Data Center Atlas (v2026.02.09) [Data set]. MSD-LIVE Data Repository. https://doi.org/10.57931/3017294